

Winterborn Restock System

Development Brief, Document 3: Scope Reconciliation

Internal document for the build team. Not client-facing.

Reconciles: the original development brief (Document 2) against a later scope review.

Authority order: where the two sources conflict, the later scope review takes precedence. Where Document 2 is silent or the two agree, Document 2 stands. Nothing here overrides the technical facts established by real client data or the proven safety design in the catalog migration decision record.

This is a scope document, mirroring Document 2's own role. It answers what changes and why. For implementation detail see `docs/superpowers/specs/2026-08-19-winterborn-restock-system-design.md` and the six build plans under `docs/superpowers/plans/`.

1. What has not changed

The later scope review independently re-derives the same core architecture Document 2 already specified, in places almost identically. This is a strong signal the underlying design was right, not a reason to redesign it. The following stand exactly as written in Document 2:

- **Inventory is a derivation, never a stored number.** The system must maintain an inventory movement ledger rather than a single global stock figure. This is the exact architecture already built.
- **Location-level inventory, not company-wide totals.** Stated as the single most important design principle in both sources.
- **Two-tier colour model.** The till sees a family only ("Blue"), the warehouse holds the full colour detail. Both sources describe this identically, down to the same worked example of several blue shades collapsing to one till option.
- **Installable web app, not native apps.** Confirmed in both.
- **No inventory writes into Square this season.** Square remains the sales record, read via webhook and reconciliation poll. Both sources describe the same sync flow.
- **Catalog restructuring protocol.** Prototype first, preserve the item's identity on write, the mid-September catalog freeze. Untouched. Already built and proven against a Square sandbox account.
- **No colour-level demand prediction this season.** Last season's data carries colour on under 5% of revenue. This stays out of scope regardless (see Section 4 on why the new restock-suggestion item does not conflict with this).
- **All six catalog write-safety guards:** automatic backup before any change, a strict allow-list of what can be touched, no deletion permitted at the code level, a hard stop on the first failure, a tested rollback, and a pre-flight check before anything runs. Untouched.

2. Reconciliation summary

Area	Document 2	Later scope review	Decision
Warehouse intake, replacing the current warehouse tool	Deferred, targeted for late in the season	Described as a core daily workflow, priced as needed sooner rather than later	MOVED EARLIER
Automated low-stock alerting	Explicit: thresholds exist, but alerts stay manual-review only until later	The system should flag proactively rather than waiting for someone to check	MOVED EARLIER (detection layer, see 3.2)
Restock quantity suggestion	Not specified; requests carry only the requested quantity	Recommend a quantity from historical velocity, current stock, and remaining season days	MOVED EARLIER , style level only, see 3.3
Overallocation across markets	Not specified	Cap a request when fulfilling it in full would starve another location	MOVED EARLIER
Shipment consolidation	Not specified	Bundle nearby restock needs into one dispatch run	MOVED EARLIER
Roles	Four: Owner, Warehouse, Market Manager, Operator	Five: splits Warehouse into a manager and an operator tier, adds a Square-only sales role	MOVED EARLIER
Dashboard status	Low-stock list, decision queue	Explicit Healthy / Low / Critical / Out-of-Stock labelling	MOVED EARLIER , labelling on existing data
QR box tracking	Core to dispatch: one label per box, scanned at the warehouse door	Made optional. Confirm accuracy first using shipment records, location inventory, and Square sales alone; introduce scanning later only if needed	SOFTENED , see Section 4
Physical warehouse tiers	Hard constraint: all discipline lives at one warehouse	Describes an optional intermediate storage tier between the main warehouse and a market, noted as not needed everywhere	NEEDS ONE FACT , see Section 5
Image recognition at intake	Not specified	Noted as a possible later addition, not needed now	STAYS LATER

New-market forecasting from external data	Not specified	Listed as a possible future input for markets with no history	STAYS LATER , no data pipeline exists; premature
Write-off flow, season-close reconciliation	Deferred	Not mentioned	UNCHANGED

3. Why these move earlier safely

None of the items below need new architecture. Each is additive to what already exists.

3.1 Warehouse intake

This is the priority item on this document, not a background line.

Scope: open the app, select "Receive Inventory," optionally take a photo, select the product and its attributes, enter a quantity, confirm. If the product does not already exist, an authorised user creates it, using the same controlled-vocabulary check already designed into the colour model, so no new mechanism is needed, only a screen.

Not included in this pass: automatic photo-based product recognition. Treated as a later addition by design.

3.2 Automated low-stock alerting

The detection half of this already exists: a restock request drafts itself automatically the moment stock crosses a threshold, deduplicated so a busy weekend cannot flood the queue. What stayed withheld until now was surfacing it clearly, rather than requiring someone to check a dashboard. Moving this earlier means the dashboard leads with what is flagged, by default.

Open question, not a blocker: sending an alert outside the app (email, text) needs a delivery channel to be chosen. Detection and on-screen surfacing move earlier now; off-screen delivery is scoped once a channel is picked, so it is not promised before it can honestly be built.

3.3 Restock quantity suggestion

More of this already exists than it first appears. Last season's real weekly sell-through per product per location is already computed, and deliberately stops at the product-style level, never colour, for the same reason colour prediction stays out of scope: last season's colour data covers under 5% of revenue, and building on it would be building on too little.

The requested feature's own worked example is style level too, no colour involved. So the version that ships now is: when a request is raised, show a system-recommended quantity next to the requested one, computed from that same historical data plus current stock and the remaining days in the season. This surfaces data already derived; it does not add a new prediction engine.

Not included in this pass: cross-market historical comparison, forecasting for brand-new markets with no history, or tuned safety margins. Those need more than one season of data.

3.4 Overallocation guard

When evaluating an open request, sum every other open request against the same warehouse stock and flag, not silently cap, if fulfilling this one in full would leave another location short.

3.5 Shipment consolidation

At pack time, show other open or soon-due requests for the same destination so a box can be combined rather than sent twice. A view on existing data, not a new subsystem.

3.6 Roles

Split the single warehouse role into a manager tier (create shipments, adjust stock, approve requests) and an operator tier (receive, pack, cannot create products or touch the catalog). Add a sales-only role for staff who only need the till, never the app, so the smallest-privilege principle already agreed holds for every role, not only some.

3.7 Dashboard status zones

Label existing threshold data as Healthy, Low, Critical, or Out of Stock rather than a bare number. No new computation, a presentation layer on data already derived.

4. What softened

4.1 QR box tracking: required to optional

Verified directly against the build: dispatching a box already posts the movement to the ledger on its own, and has never required a scan to run. The scan screen is a convenience for finding a box quickly, layered on top of the underlying movement record, not load-bearing beneath it. No rework is needed. The system already meets the stated bar: shipment records, location inventory, and Square sales are the mechanism; scanning is optional confirmation on top.

5. The one item that needs a fact, not a decision

5.1 Does any market operate a second storage tier today

The single-warehouse rule in Document 2 was a specific, verified observation about how this business's markets actually run: informal, staff drawing stock freely, nothing counted on site. The later scope review's optional intermediate tier (main warehouse to a market's own storage to the selling point) is explicitly hedged as not needed everywhere.

Checked against the data model: this does not require a new design. A location is already either a market or a warehouse, and a stock transfer already takes any two locations as its endpoints; nothing assumes exactly one warehouse exists. A market's local storage, if one is real, is simply a second warehouse-type location, and main warehouse to that location to the market's till is two ordinary transfers, not a new mechanism.

So this is already designed for, not a build item. The only reason it is inactive today is that no second real storage location has been confirmed. The one fact that resolves this: does any market run physical storage of its own right now, or is this planning ahead for growth. If real, it is a data entry, no code change. If not yet real, it stays exactly as designed and gets activated when it is.

6. What this document deliberately leaves alone

- The mid-September catalog freeze and the pilot market target date, unchanged
- The single catalog write window before the season, and every safety guard around it, unchanged
- No colour-level prediction, this season or next, until a full season of family-level sell-through exists to build on
- Write-off flow and season-close reconciliation, deferred, not mentioned in the later review, no reason to move
- Photo-based product recognition, deferred by the same review that raised it

Prepared by the build team, reconciling Document 2 against a later scope review under instruction that the later review carries precedence on scope questions where the two disagree.